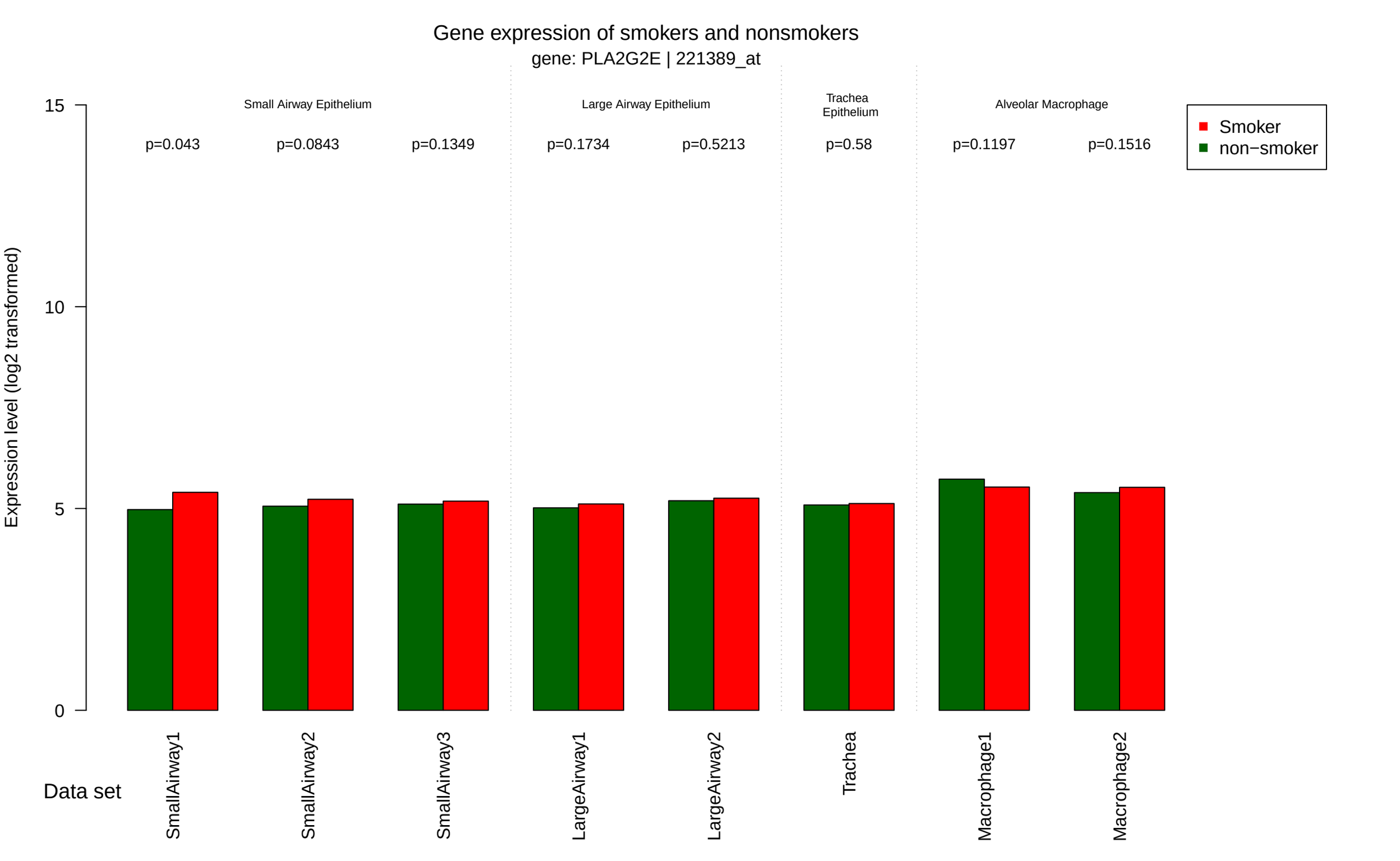
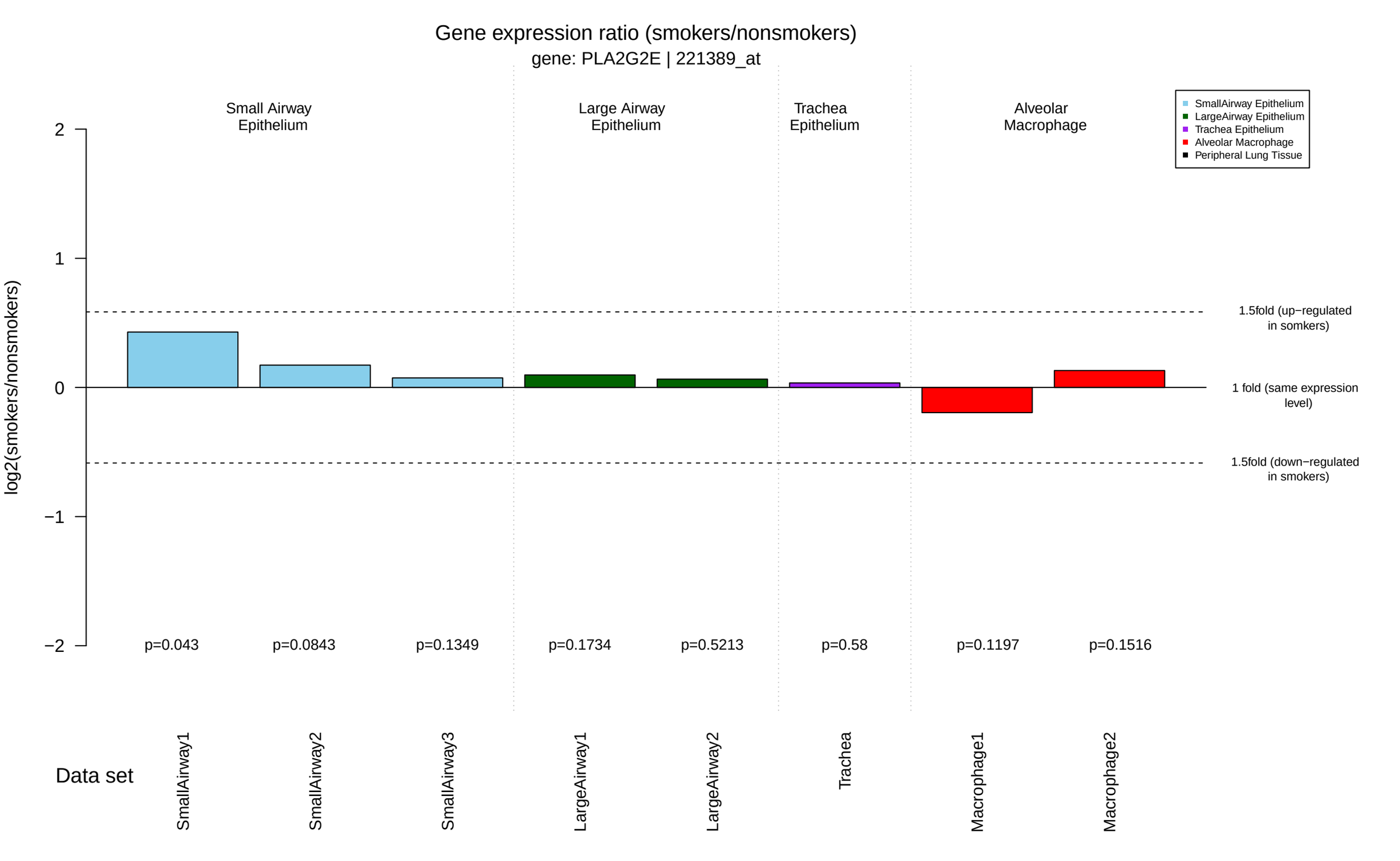
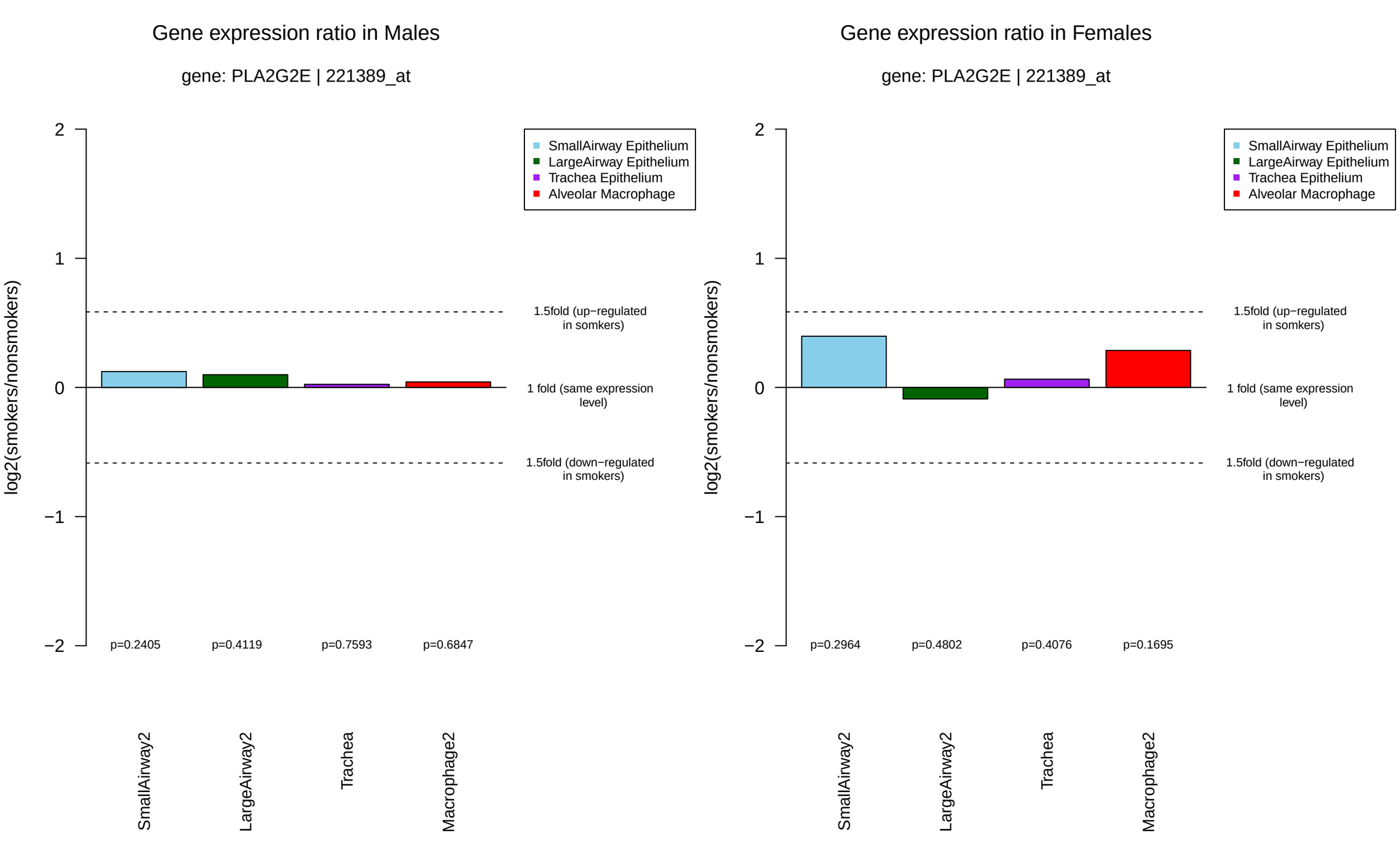
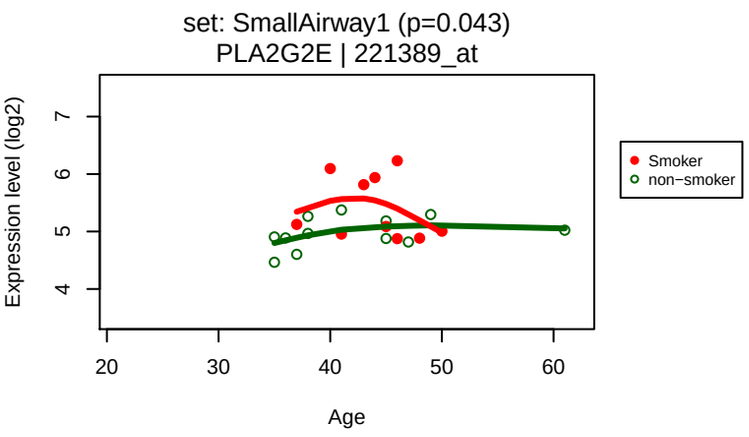
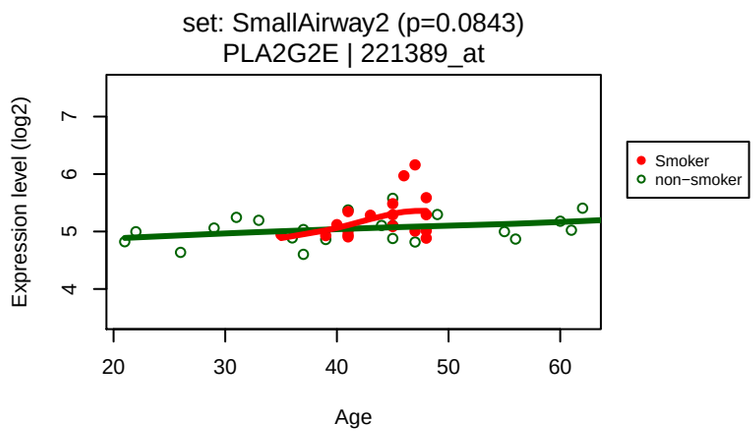
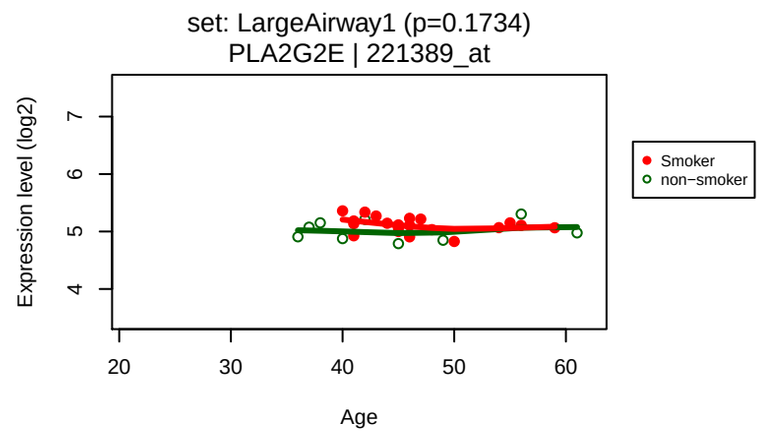

Gene expression of smokers and nonsmokers
gene: PLA2G2E | 221389_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.043
p=0.0843
p=0.1349
p=0.1734
p=0.5213
p=0.58
p=0.1197
p=0.1516
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

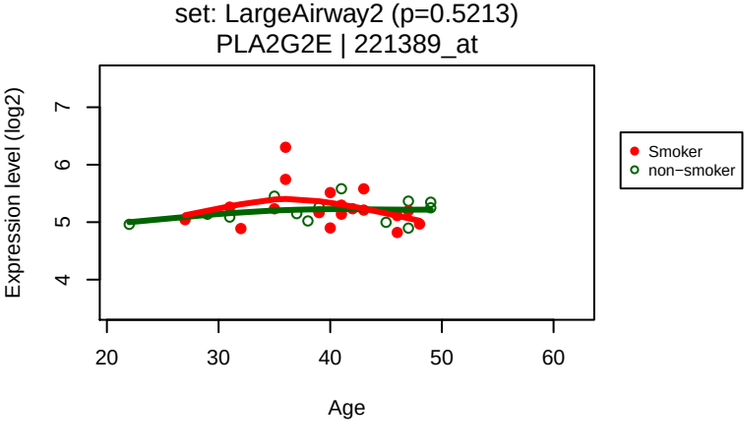

Gene expression ratio (smokers/nonsmokers)
gene: PLA2G2E | 221389_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.043
p=0.0843
p=0.1349
p=0.1734
p=0.5213
p=0.58
p=0.1197
p=0.1516
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

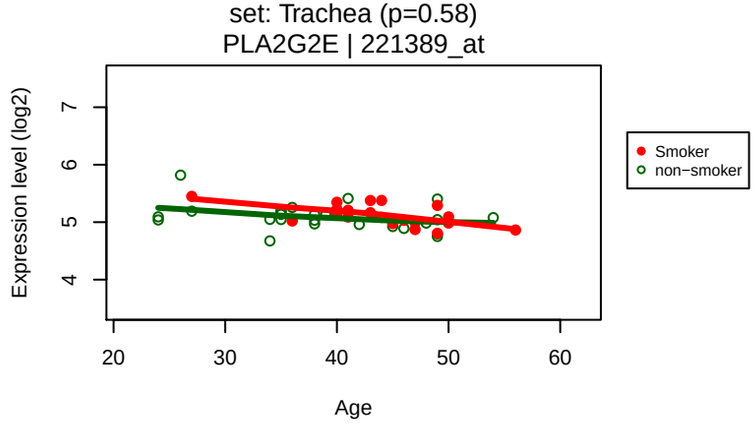

Gene expression ratio in Males
gene: PLA2G2E | 221389_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2405
p=0.4119
p=0.7593
p=0.6847
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLA2G2E | 221389_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2964
p=0.4802
p=0.4076
p=0.1695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

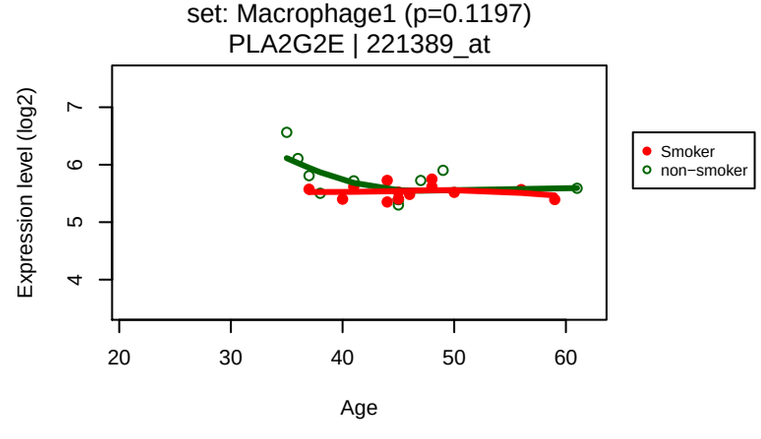

set: SmallAirway1 (p=0.043)
PLA2G2E | 221389_at
Expression level (log2)
Age
Smoker
non−smoker

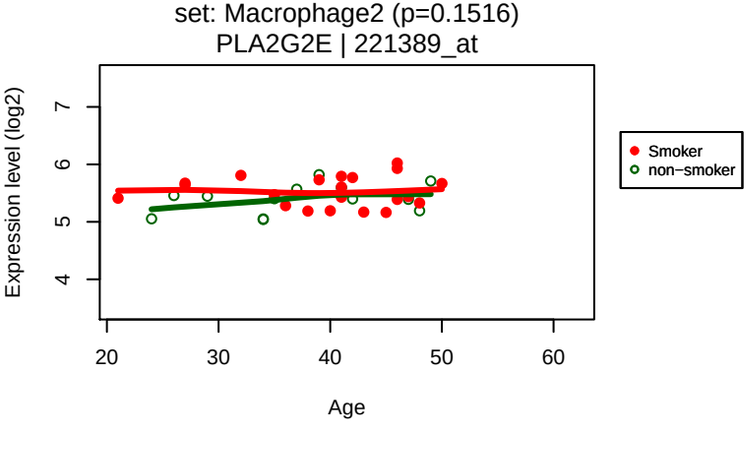

set: SmallAirway2 (p=0.0843)
PLA2G2E | 221389_at
Expression level (log2)
Age
Smoker
non−smoker

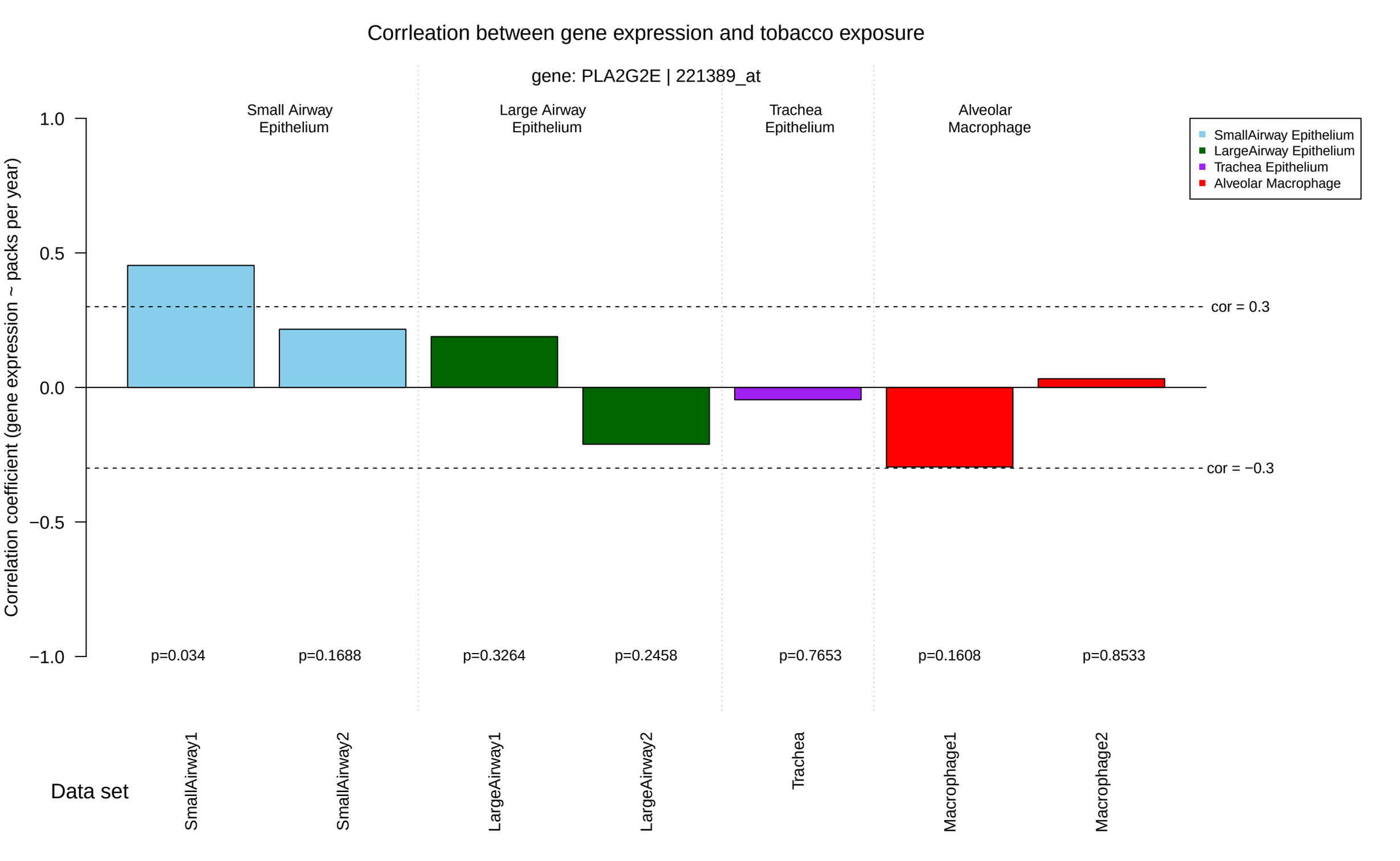

set: LargeAirway1 (p=0.1734)
PLA2G2E | 221389_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5213)
PLA2G2E | 221389_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.58)
PLA2G2E | 221389_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1197)
PLA2G2E | 221389_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1516)
PLA2G2E | 221389_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLA2G2E | 221389_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.034
p=0.1688
p=0.3264
p=0.2458
p=0.7653
p=0.1608
p=0.8533
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2